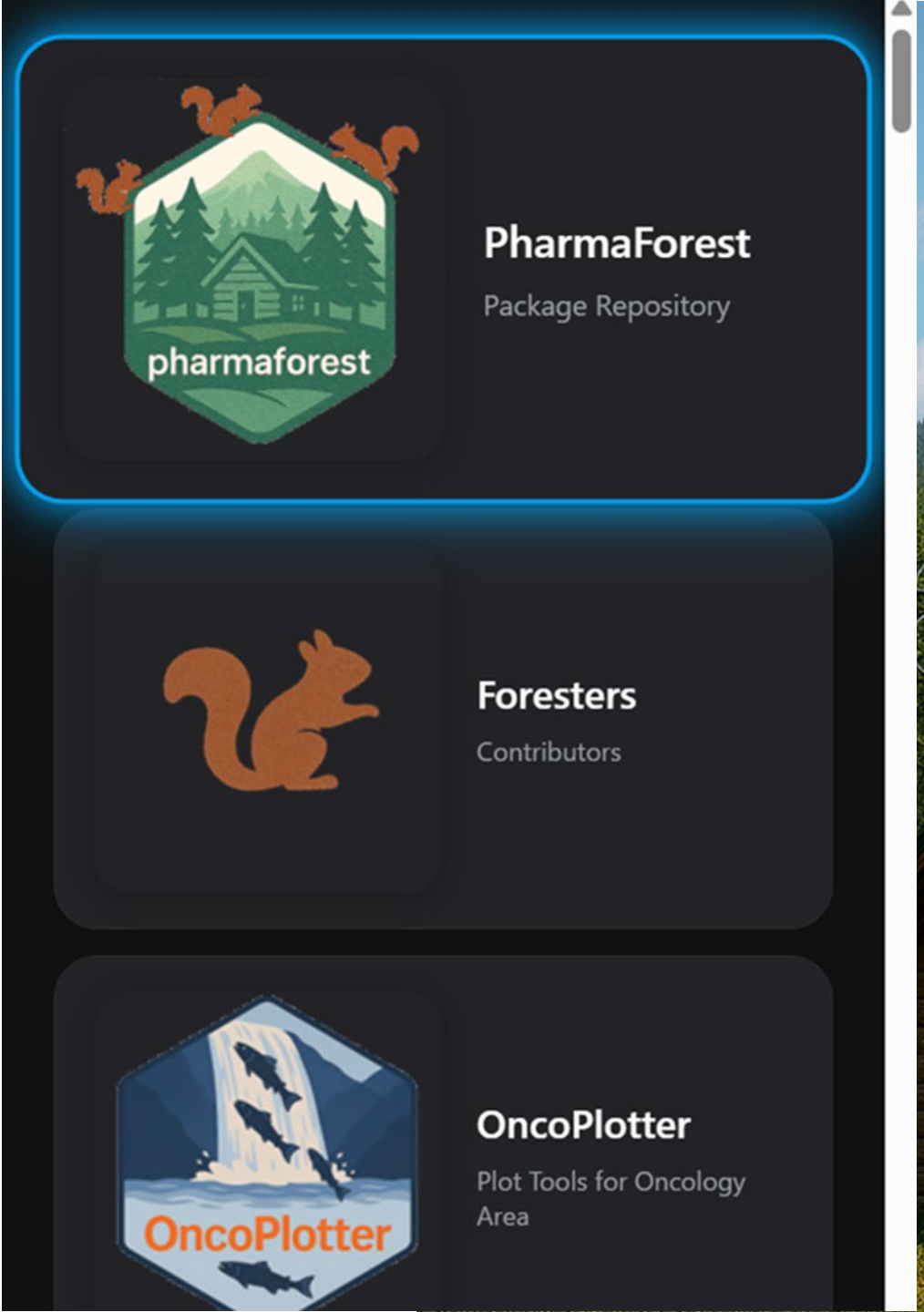


Introduction of PharmaForest – Forest of SAS packages

Ryo Nakaya, Takeda
Yutaka Morioka, EPS
Hiroki Yamanobe, Maruho

Phuse US Connect 2026 @ Austin
24March2026

Scan it!

Our team

PHUSE Japan OST Working Group	
Masashi Aonuma (Astellas Pharma Inc.)	Ippei Akiya (ICON Clinical Research GK)
Naoko Izumi (Novartis Pharma K.K)	Shunsuke Goto (Novartis Pharma K.K.)
Masashi Tanaka (EPS Corporation)	Yutaka Morioka (EPS Corporation)
Tomoyuki Namai (Chugai Pharmaceutical Co., Ltd.)	Tomoki Nishikawa (Kyowa Kirin Co., Ltd.)
Kensuke Sasaki (DATACK Co., Ltd.)	Ryo Nakaya (Takeda Pharmaceutical Co., Ltd.)
Masataka Higashijima (Eli Lilly Japan K.K.)	Kazuyuki Miyashita (Nippon Boehringer Ingelheim Co., Ltd.)
Yasutaka Moriguchi (GlaxoSmithKline K.K)	Yuichi Nakajima (GlaxoSmithKline K.K.)
Hiro Kosako (A2 Healthcare Corporation)	Hiroki Yamanobe (Maruho Co., Ltd.)

phuse·share
connect·share·advance

Scan it!

Objective

- Introduce and promote PharmaForest
- Find collaborators

Scan it!

Open-source in Pharma

- Recent trend in pharmaceutical industry to **leverage open-source software such as R and Python** is remarkable
- SAS is still in its position of center in the industry, but open-source mind is not yet flourished enough
- Potential of R and packages in statistical clinical reporting is increasingly explored and **Pharmaverse** was formed in 2020
- **AI/ML and automation** is often done by Python

Scan it!

Key factors of open-source

- License
- Standardized format to share

Scan it!

Definition of open-source

1. Free Redistribution
2. Source Code
3. Derived Works
4. Integrity of The Author's Source Code
5. No Discrimination Against Persons or Groups
6. No Discrimination Against Fields of Endeavor
7. Distribution of **License**
8. **License** Must Not Be Specific to a Product
9. **License** Must Not Restrict Other Software
10. **License** Must Be Technology-Neutral

open source initiative®

https://opensource.org/osd

Having license matters!

Scan it!

Standardized structure (package)

```
yourpackage/
├── DESCRIPTION
├── NAMESPACE
├── R/
│   ├── function1.R
│   └── function2.R
├── man/
│   ├── function1.Rd
│   └── function2.Rd
├── tests/
│   └── testthat/
├── data/
├── vignettes/
└── README.md
```

Place to share (repository)

CRAN R
The Comprehensive R Archive Network

python™ Package Index

pharmaverse

GitHub

What about SAS??

Scan it!

SAS Packages Framework(SPF)

SAS Packages Archive(SASPAC)

Pharmaforest

Yes, we have.

Scan it!

SAS Packages Framework(SPF)

- SAS Package is based on **standardized structure** defined by SAS Packages Framework(SPF)
  https://github.com/yabwon/SAS_PACKAGES
  Started in 2019 by Bartosz Jablonski*

```
<packageName>/
├── description.sas
├── license.sas
├── 001_macro/
│   └── mymacro.sas
├── 002_function/
│   └── myfunc.sas
├── 003_format/
│   └── myformat.sas
├── 004_data/
│   └── mydata.sas
├── 005_exec/
├── 006_clean/
└── 999_test/
```

sas packages

the way to share

Scan it!

SAS Packages Archive(SASPAC)

- A repository of packages(12 packages as of December, 2025)
  https://github.com/SASPAC
  by Bartosz Jablonski

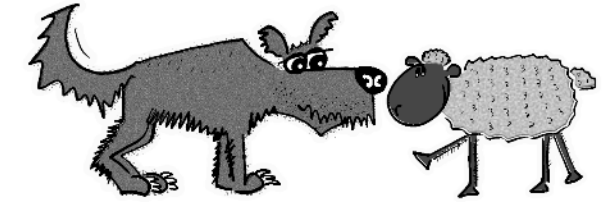
bputil Public
Basic Popular UTiLity programs
1 0 0 0 Updated on Jul 16

sasjscore Public
SASjs Macro Core library, packaged for SAS Packages
SAS 3 MIT 0 0 0 Updated on Jul 8

gigs Public
Convert between measurements and z-scores/percentiles for INTERGROWTH-21st and WHO Child Growth Standards, and classify growth patterns.
3 1 0 0 Updated on Nov 22, 2024

sasplotter Public
modern graph suite for SAS
2 0 0 0 Updated on Jul 13, 2024

macroarray Public
Macroarrays for macro codes
SAS 9 0 0 0 Updated on May 22, 2024



sas packages

the way to share

Scan it!

PharmaForest

- Collection of **SAS Packages** for pharmaceutical industry inspired by Pharmaverse project in R
- **Open-source**
- The repository has started in June, 2025
- Founded in Japan by authors of the presentation
- 43 packages
- Based on SAS Packages Framework(SPF)

Scan it!

Goal

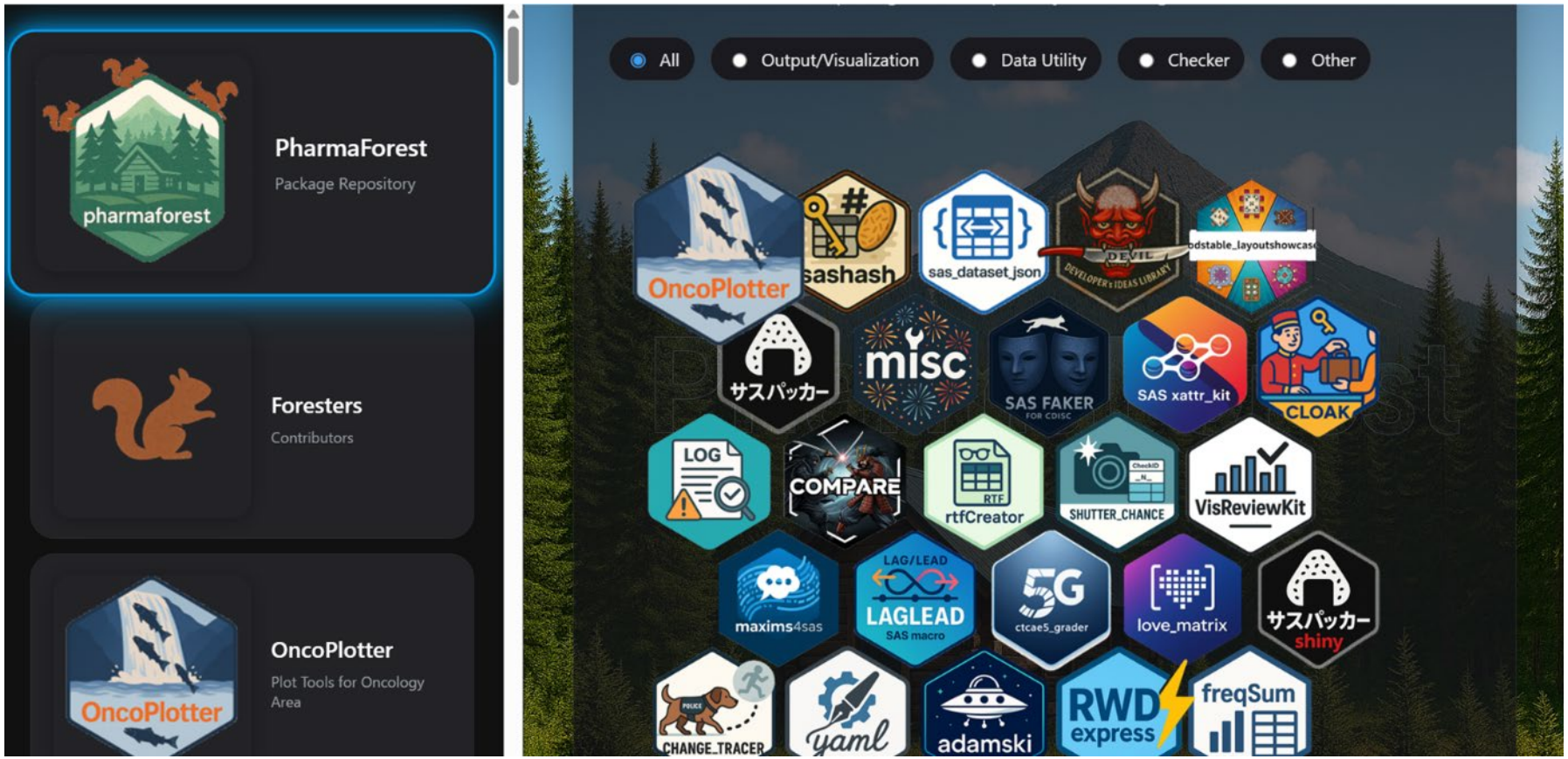
- *Through SAS Packages, we want to **<u>actively encourage sharing</u>** of SAS know-how that has often stayed within individuals. By doing this, we aim to build up collective knowledge, boost productivity, ensure quality through standardization, and energize our community.*
- https://github.com/PharmaForest

pharmaforest

Scan it!

Gallery page of PharmaForest

- https://pharmaforest.github.io/ or QR code



PharmaForest Ecosystem Map

Scan it!

Output / Visualization

OncoPlotter
VisReviewKit
freqSum
Basket MiningKit
RTF rtfCreator
odstable_layoutshowcase
trial_flow

Other

サスパッカー shiny
DEVIL
yaml
サスパッカー
misc
maxims4sas

Checker

COMPARE
SHUTTER_CHANCE
CAT_REFEREE
LOG
CHANGE_TRACER
VALIVALI

Data Utility

sashash
sas_dataset_json
SAS xattr_klt
LAG/LEAD LAGLEAD SAS macro
love_matrix
RWD express
datacut_sas
SAS FAKER FOR CDISC
CLOAK
5G ctcae5_grader
adamski
SASToolbox

As of Nov2025

SASPACer package

- Support creating SAS Packages

Scan it!

https://github.com/PharmaForest/SASPACer

Type	Package
Package	testPackage
Title	My first SAS package
Version	0.0.1
Author	John Smith(john.smith@mail.com)
Maintainer	John Smith(john.smith@mail.com)
License	MIT
Encoding	UTF8
Required	"Base SAS Software"
ReqPackage	"Baseplus (2.1.0)"
Description	## The myPackage ## The `myPackage` is my first SAS package. ### References ### 1. Bartosz Jablonski. "My First SAS Package - a How To". SGF

description | license | 01_libname | 02_forma

name	help	body	location
mcrone	This is mcrOne macro. (No need to write location column if content is written in body column.)	%macro mcrOne(); %put **Hi! This is macro &sysmacroname.**; data _null_; set myLib.smallDataset; p = f1(n); p + f2(n); put (n p) (= fmtNum.); run; %mend mcrOne;	
mcrtwo	This is mcrTwo macro. (No need to write body column if content is in a file written in location column, SASPACer reads the file.)		C:¥temp¥addcnt¥mcrtwo.sas

02_formats | 03_functions | 04_data | 05_lazydata | 06_macro | 07_test

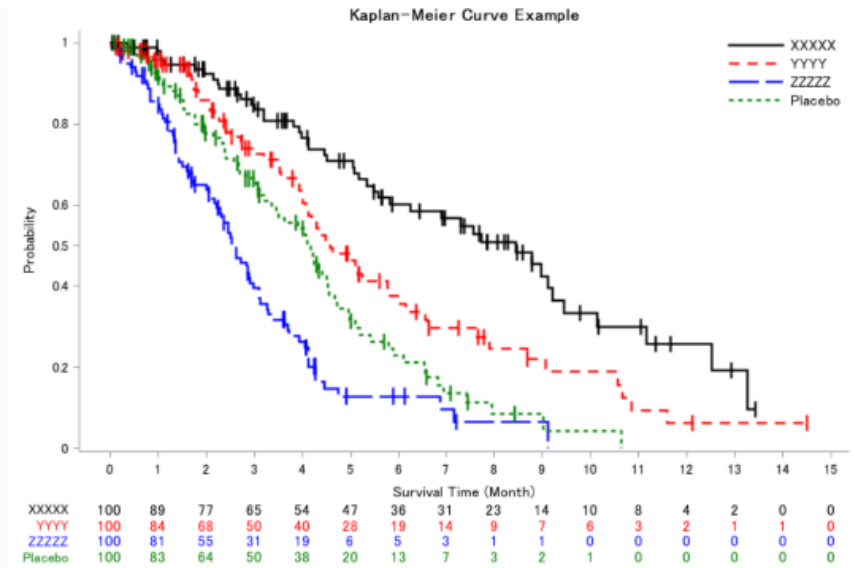
```
%installPackage(SASPACer, mirror=pharmaforest)
%loadPackage(SASPACer)

%ex2pac(
    excel_file              = /path/to/excel/excelfile.xlsx,
    package_location        = /path/to/output,
    complete_generation = Y     /* Y: Run %generatePackage() */
)
```

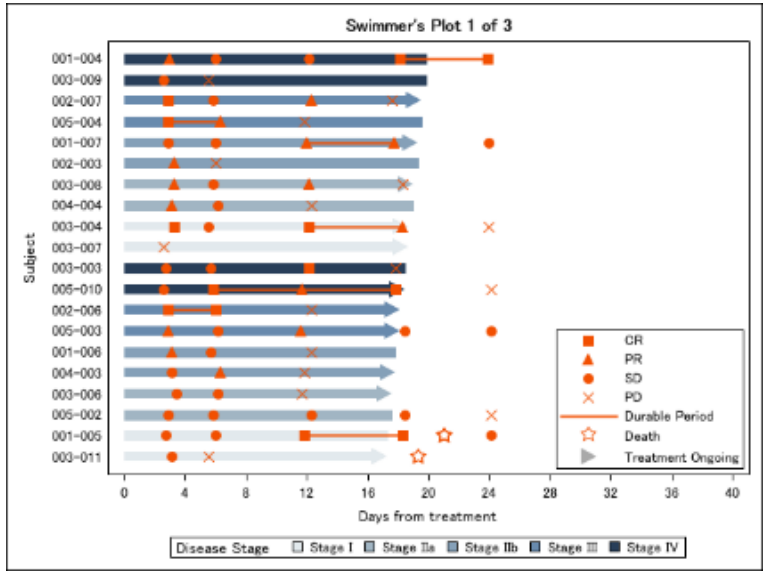
OncoPlotter package

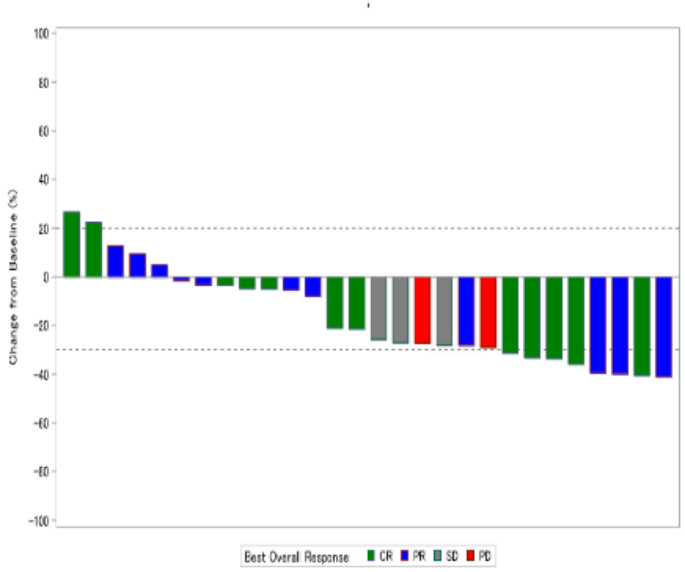
- Plotter for Oncology figures

Scan it!

https://github.com/PharmaForest/OncoPlotter








```
%installPackage(OncoPlotter, mirror=pharmaforest)
%loadPackage(OncoPlotter)
```

Adamski package

- CDISC ADaM creation tool

```
%installPackage(adamski, mirror=pharmaforest)
%loadPackage(adamski)

data have;
  do SEX="F","M"; do AGE=12 to 17; output; end; end;
run;

data want;
  set have;

  /* With condition */
  %derive_var_merged_exist_flag(
              dataset_add = SASHELP.CLASS,
              by_vars     = AGE,
              condition   = %nrbquote( SEX = "F"),
              new_var     = AGE_IN_CLASS_F_ONLY,
              true_value  = Yes,
              false_value = No
  );
run;
```

Scan it!

https://github.com/PharmaForest/adamski

Aiming to create similar functionality to admiral in R and add original macros

SAS_dataset_json package

Scan it!

https://github.com/PharmaForest/sas_dataset_json

- Dataset JSON

```
%installPackage(sas_dataset_json, mirror=pharmaforest)
%loadPackage(sas_dataset_json)
proc datasets nolist;
  modify adsl;
  xattr add ds originator="X corp."
      fileOID="www.cdisc.org/StudyMSGv2/1/Define-XML_2.1.0/2024-11-11/"
      studyOID="XX001-001"
      metaDataVersionOID="MDV.MSGv2.0.SDTMIG.3.4.SDTM.2.0"
      sourceSystem_name="SASxxxx"
      sourceSystem_version="9.4xxxx";
  xattr add var
      STUDYID (label="Study Identifier"
                dataType="string"
                length=8
                keySequence=1)
      USUBJID (label="Unique Subject Identifier"
                dataType="string"
                length=7
                keySequence=2)
      TRTSDT (label="Date of First Exposure to Treatment"
                dataType="date"
                targetDataType="integer"
                displayFormat="E8601DA.") ;
  ;
run;
quit;
%m_sas_to_json1_1(
      outpath = /project/json_out,
      library = WORK,
      dataset = adsl,
      pretty = Y
);
```

Yaml_writer package

Scan it!

https://github.com/PharmaForest/yaml_writer

- Writing yaml in SAS

```
%installPackage(yaml_writer, mirror=pharmaforest)
%loadPackage(yaml_writer)

%yaml_start(outpath=C:/Temp, file=sample);
#This YAML is a sample &nw.;
all: &nw.;
  - name: MHPRESP &nw.;
    operator: equal_to &nw.;
    value: "Y" &nw.;
  - name: MHOCCUR &nw.;
    operator: equal_to &nw.;
    value: "N" &nw.;
%yaml_end();
```

test.yaml

C: > Temp > test.yaml

```yaml
all:
  - name: MHPRESP
    operator: equal_to
    value: "Y"
  - name: MHOCCUR
    operator: equal_to
    value: "N"
```

Also have %dataset_export() for exporting data to yaml

Forest Navigators (GenAI-powered)

Scan it!

Dr. Forest

(Master Navigator,

General, #1-#15 packages)

Dr. Apple

(Support Navigator,

#16-#30 packages)

Rio

(Support Navigator,

#31- packages)

Oba-chan

(SAS package lady,
a SAS package cook)

https://github.com/PharmaForest

Scan it!

Conclusion

- PharmaForest was built to encourage SAS programmers to share their knowledge in a unified format of SAS Packages.
- We look forward to hearing from anyone from the world who are interested.

Let’s create open-source community in SAS together!

Scan it!

References

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